

DESIGN AND IMPLEMENTATION OF A AUDIO CONTROLLED REMOTE TOY CAR

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Abstract: This document highlights the design and implementation of an audi controlled remote toy car . The engineering design procedure is used as the bedrock for the structure of this report . The engineering design process is highlighted by the analysis of the problem to be solved.Brainstorm of the possible solutions that can be implemented . Exploration of established ideas within the scope of the problem at hand,in hopes of increasing the design quality of the solution. Establishment of the constraints and the criteria for the project outline,a design of the proposal to meet those constraints and criteria and testing and evaluation of the proposed solution.

Key words: DTMF , audio controlled signal , tone - detectors , dual tone signal

1. INTRODUCTION

This report is a catalogue of the information collected in pursuing a solution to the headlined topic.It showcases the steps taken to formulate the final design and the critique of the realised solution .

2. BACKGROUND

2.1. Project Specifications

The aim of the project was to use an audio signal to control a remote car , with the link being a wireless connection. .The final design was to be tested in an environment where other drivers would be testing their own designs , making immunity very imperative to the final design.

2.2. Constraints

The design constraints that were proposed for this implementation included the control signal limit of 20Hz to 20kHz . A maximum of 10 volts (V) , to be supplied for the circuit , with only a maximum of 5 V to be across the DC motors . There was to be no use of any micro controllers in the design and no modifications were to be made to the chassis that was issued out.

2.3. Assumptions

Toward the testing procedure and track the team made assumptions being , there would be good weather on the testing day. The designed solution would be totally immune and work with no problems, Smooth simple track to manoeuvre with no complex twists of turns.

2.4. Literature Review

The control of a remote to car can be achieved in a number of ways each depending on the use case. The author in the article [1] , achieves the control of the remote car using a Wi-fi network module ESP8266 paired

with Arduino-Uno to relay the directional command system . Huang Jian [2] manages to design a bluetooth based design scheme through the use of a HC-05 module. The design is based on an intelligent car where functionality such as real time monitoring in terms of tracking , temperature and humidity detection and wireless control can be achieved and are necessary. Yajing Yu [3] , highlights a different approach for the intelligent remote toy car using a STC89C2 microcontroller as the core control with three other subsystems for the realisation of sound control , tracking and obstacle avoidance with adjustable traversal speed . Gupta et al [4] , shows a DTMF based control scheme using a CM8870 to output a 4 bit binary output that allows for a very simple control scheme , and they manage to eliminate a micro controller in that implementation.

The above solutions show the range of solutions that are normally pursued when looking for a solution to the implementation for the control of a remote toy car.It does present a gap in the knowledge of implementing the solution using audio signals as most solutions are through the use of Wi-Fi modules , bluetooth modules and radio frequency transmitters . This paper explores the simpler methods of control , as highlighted by the project constraints , eliminating the use of microcontrollers and necessitating the need of an audible signal for control.

2.5. Possible Solutions

The provided reports from the previous years students were analysed first raising interesting solution choices . These solutions however were made obsolete due to a minor additional constraint of the use of no micro controllers for the overall design that was introduced later during the design window.. This change resulted in the team brainstorming to find new solutions that met the new constraints .

On the 22nd of March , the group held a design meeting to analyse different ideas in search for new viable design

solutions. The objective was to raise two different approaches with one being a simple implementation and the other being a complex implementation. Two solutions explored simultaneously would ensure a higher chance of success in finding a solution that works. The simpler solution would eliminate immunity and work based on the principle of minimum viable product.

The first idea proposed was to use analogue filters and decoders. This solution would be implemented in a manner of cascading bandpass filters each tuned to a specific frequency range and corresponding to the control signals i.e 15kHz to 20kHz. Op-amps were to be used in this solution to create the filters as they proved easier to build and design due to online tools i.e Filter Design Tool by Analogue Devices.

Once the incoming audio signal was though it would be taken to the analogue detectors such as frequency to voltage converters devices like the LM2907 to convert the filtered audio frequencies into their appropriate voltage levels, as these integrated circuits (ICs) can provide a DC output they would be paired with comparators to direct specific logic highs to the L298N motor driver to drive the toy car. The other analogue detector that was considered was the tone detectors such as the LM567 or the NE567, this IC allows for the detection of a calculated frequency when used in the configuration highlighted by figure 1. When the desired signal is detected a logic low is produced at the output pin 8, this logic low was to be used to control the L298N by using it in conjunction with a NOR gate IC, the 74LS02, turning the logic lows into logic highs. This configuration is used as suggested by Texas Instruments datasheet [5].

The second idea that was proposed was the use of modulation techniques mainly being the Frequency Shifting Keying (FSK) or the Amplitude Shift Keying (ASK). The FSK solution was based on having two distinct audio frequencies to represent the binary data i.e 1KHz for '0' and 2kHz '1', the encoded signal would be received on the other side and a FSK demodulator circuit was to be built and once decoded the use of Frequency to Voltage Converter (FVC) was to be used to overlay the command to the L298N motor driver. Comparators would be used to direct the logic highs of the FVC, to the specific commands.

The ASK solution was proposed to use only one audio frequency solution, that was to be used with a ASK modulator circuit for the transmission using a voltage controlled amplifier or analogue multiplier i.e AD633. The aim was to vary the amplitude of the carrier signal based on the control commands, and then using an envelope detector circuit the amplitude variations were to be extracted and comparators with thresholds used to decode the command scheme.

The solutions that were to be pursued initially were the FSK and the analogue filter solutions, but as the original ideas were based on a shallow understanding signal modulation. It was later decided to drop the FSK option as it proved to be too complex to grasp. In its place the idea incorporating the tone detectors without the filter system was chosen, the filter was abandoned due to high rejection of out of band signals and noise immunity features on the LM567.

3. DESIGN PROCEDURE

As two solutions were pursued simultaneously so as to increase the odds of a solution being found, two members were selected to be a leader in their 'mini' project with the third member acting as a freelance member shifting between both solutions. The third member would help with anything the project leader needed ranging from building the circuit, research on component structure etc. During development the first solution proved to be problematic due to the components not behaving as simulated, hence more attention was given to the tone detector solution over time. The first solution was only dropped a week before the final presentation week, as there were still quality assurance things to be sorted out with the second solution.

Splitting up the teams in this manner proved to be quite advantageous, as other groups ended up shifting to other solutions not initially proposed and ended up putting themselves in time crunches to rush to solutions with some unfortunately not finding a solution quick enough. Table 1, shows the overall division of work within the group highlighting the contribution of each member. With figure 6 showing the Gantt Chart for the weekly progression of the project. Using the circuit structure highlighted by Texas Instruments in the 567 datasheet [5], along with the equation highlighted in the appendix A, the frequencies for the control commands were selected. A NOR gate IC was first used to turn the logic lows into logic highs, given that two logic lows give one logic high this was excellent as it allowed for the use of a dual tone signal of two frequencies guaranteeing the system against interference from other teams. The interface with the L298 motor driver was direct from the NOR gate, to the input pins 1, 2, 3 and 4. Figure 3 in the appendix shows the block diagram for the realised solution.

3.2 L398 Motor Driver

The direct interface with the motor driver provided a super simple method of controlling the remote controlled car, with a variable Pulse Width Modulation circuit prepared as the wheel of the remote car proved to be moving at different speeds causing it to bank to the right when required to move straight. Due to time constraints and some slow days in development it was proposed to not figure out a system to control the car by turning left and right while going forward instead by going in reverse

the same outcome could be achieved . Figure 2 highlights the schematic of the motor driver

3.3 Sampling the control signal

Initially a simple electret microphone was used but this produced sampling problems as the LM567 operated with minimum amplitude requirements of only 75mV. This is while the electret mic has a maximum amplitude of 20mV. As a result a preamplifier circuit would be needed for the mic , but regardless of simulation and building the circuit did not work. The KY08 sound module was later used but also presented the same problems unless you were very very close to the mic .

As this method of control was undesirable since the team wanted a long range of control , a MAX9814 microphone module was purchased from a local electronics supplier as it had a maximum of 2Vpp amplitude with various gain configuration ranging from 20dB , 40 dB and 60dB gain .Upon installation the mic proved to work just fine , with a maximum control distance of about 1 metre away with a loud enough speaker and a tone around 3kHz to 6kHz.

3.4 Schematic for Control

Figure 3 highlights the control scheme that was used to control the remote car . An app was designed for the ease of control for the remote car with the interface of the application shown in figure 4 . A total of four tone detectors were used each set to a specific frequency , with detector A being set to 1937 Hz , detector B set to 3635Hz and 4564 Hz for tone detector C and 7133 Hz for the tone detector D. The tone detectors were put into pairs for control, namely A and B for moving forward and this logic high was put into input pin 1 and 3 of the L298 . A and C were used for the right command with the logic high being put into input pin 4 and for left turn B and C were used with the logic high being put into input pin 3. The tone detector D was discarded from the control scheme as high frequency tones produced harmonics that had a higher amplitude from the initially expected tone making the designed NOR gate system inefficient . Since the first three tone detectors could be used without any problems their combinations were used in the control scheme .

4. ANALYSIS OF THE FINAL DESIGN

During the testing day the final design solution was presented and it worked as expected ,without any major issues and completed the full course with a time of 7.38 minutes , which the team felt was a bit too long and could definitely be worked on in any future considerations .Before testing there were attempts to increase the speed of the remote car by using two power banks in series to form a 10V source to the motor driver but the car became jittery and the motion was not smooth . A XL600L DC-DC converter module was also used to boost the

speed of the remote car but was scrapped later due to inconsistencies of operation as the DC motors sometimes would run then at other times would not run at all . This resulted in the removal of the XL600L boost converter module . NOR gate IC , was later removed as it gave inconsistent control due to current imbalance and custom NOR gates were built , using the circuit in figure 5.

4.2 Design Improvements

In hindsight for future design implementations , more thought could have been put into the control scheme that allowed the driver to turn while going forward as the turning while reversing motion confused the driver at times due to orientation . This happened because the driver did not get enough time to test drive the designed solution as planned and went to the track slightly under prepared. The solution implemented proved to be a bit costly, table 2 highlights the total costs incurred in making the design. More alternative solutions can be approached to cut costs such as a XE2211 , as it has two logic outputs per IC. The L298 used too much current , a custom H-Bridge using mosfets could have been built and faster speeds could have been achieved as a FET H-bridge has lower voltage and current drops .

CONCLUSION

There was room for improvement in the presented solution that would have allowed for better performance on the track . Regardless the final solution met all constraints and worked as intended which is to be considered a successful implementation

REFERENCES

- [1] Joni Welman Simatupnag , Michael Yosua , “ A Remote Controlled Car Using Wireless Technology “ , Journal of Electrical and Electronics in Engineering , December 2016 , Last Accessed 14 May 2024
- [2] Huang Jian , “Design of Bluetooth Remote Control Intelligent Car “ , Advances In Computer Science Research , Volume 82 , XiJing University , China , January 2017, Last Accessed 14 May 2024
- [3] Yajing Yu , “The Implementation of a Mobile Intelligent Mobile Toy Car “, EDP Sciences , 2018 , School of Management , Shanghai University , China Last Accessed 14 May 2024
- [4] Sabuj Das Gupta , Arman Riaz Ochi , Mohammad Sakid Hossain , Nahid Alam Siddique . “Designing & Implementation of Mobile Operated Toy Car by DTMF “,International Journal of Science and Research Publications , Volume 3 , Issue 1 , January 2013 , Last Accessed 13 May 2024
- [5] Texas Instruments LM567x Tone Detector , Revised January 2022 , Available at : <https://www.ti.com/lit/ds/symlink/lm567c.pdf?ts=1715631828997> , Last Accessed 14 May 2024

APPENDIX

APPENDIX A : FORMULAE

Below is the equation used to set the control frequencies :

$$f_o = \frac{1}{1.1 * R1 * C1}$$

Given that Texas Instruments provided all the necessary capacitor values , only the value of resistor R1 which bridges between pin 5 and 6 on the IC needed to be calculated . The value of capacitor C1 was given to be 0.1uF. C2 was given as 1uF and C3 as 2.2uF.

APPENDIX B : FIGURES AND TABLES

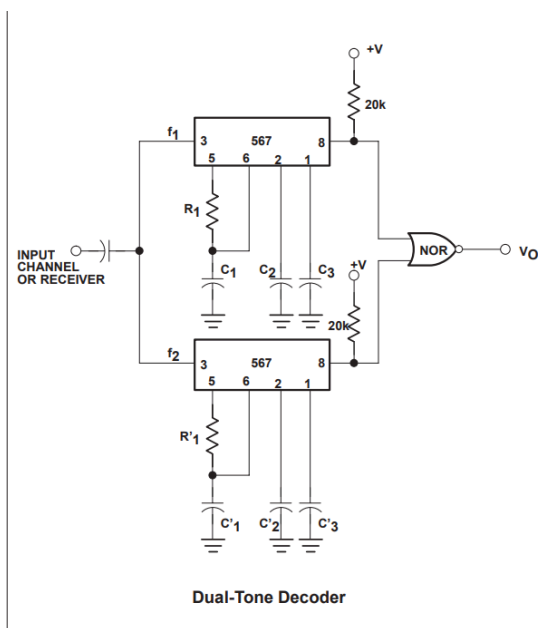


Figure 1 : Touch Tone Detection Configuration from NE567 Philips Datasheet

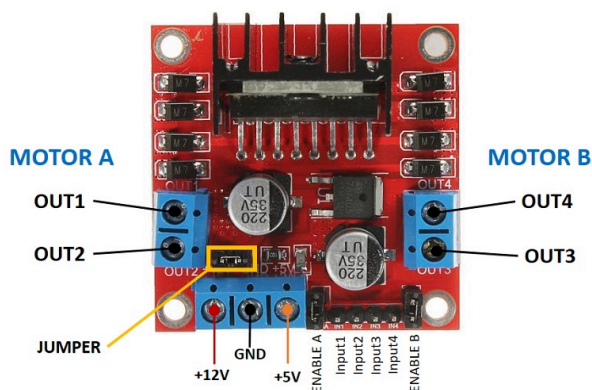


Figure 2 : Pin out for the L298N motor driver

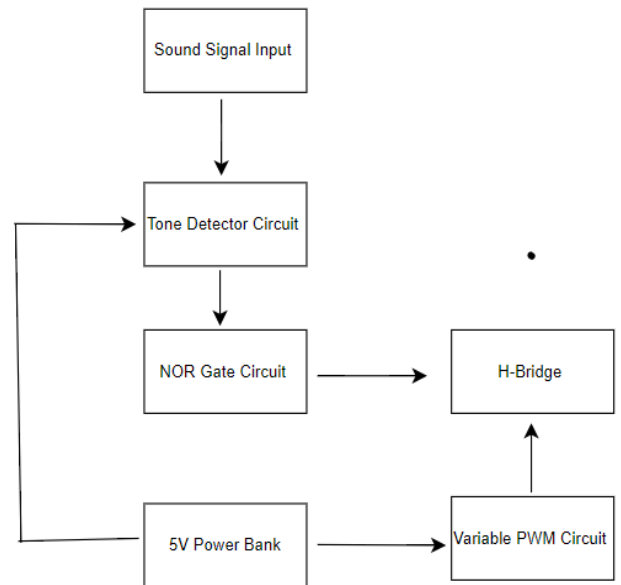


Figure 3 :Block diagram for the circuit configuration



Figure 4 : Interface for control of the remote car from the built application

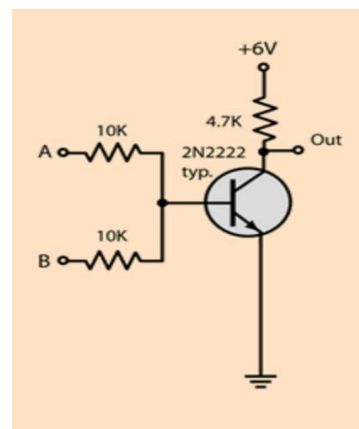


Figure 5 : NOR gate circuit to substitute the 74SL02

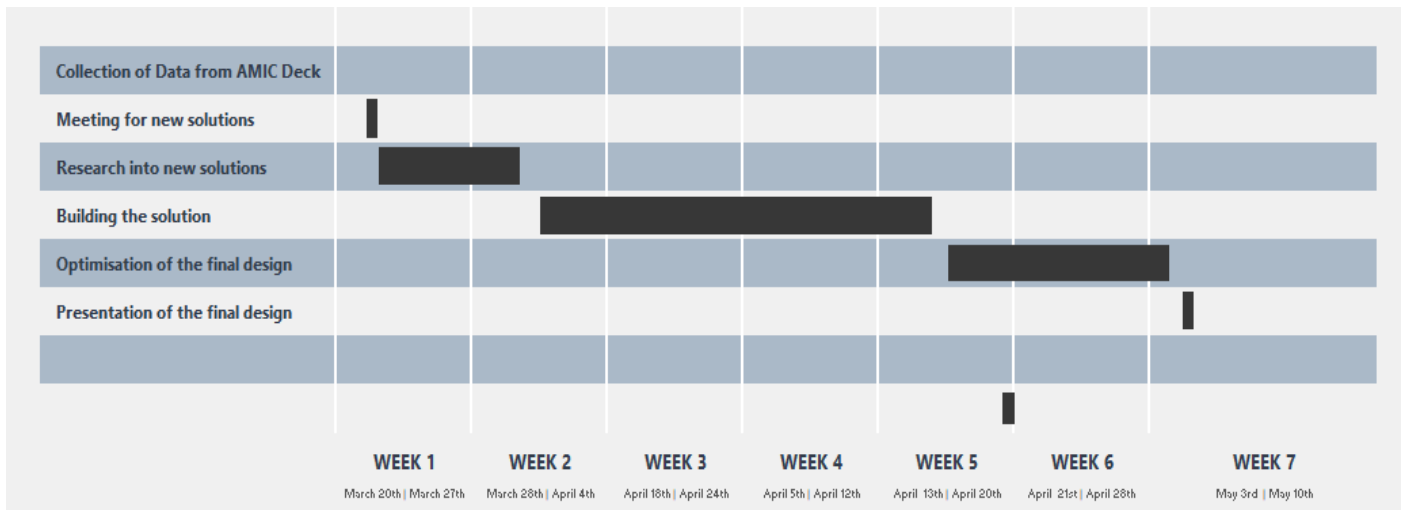


Figure 6 : Gantt Chart for project progression

Table 1 : Division of Work

	Engineer 1	Engineer 2	Engineer 3
Research of viable Solutions	✓	✓	✓
Development of Solution	✓	✓	✓
Building the Circuit	✓	✓	✓
App Development		✓	
PCB Design			✓
Driving the remote control car		✓	

Table 2 : Total Cost Incurred during development

Item	Costs
LM567	4 x 30
MAX9814	2 x 104
XL600L DC-DC Boost Converter	140
Power Banks	470
Total Cost	930

Table 3 : SWAT Analysis for the final design implementation

Strengths	Weakness
1. Implemented in the most simplest way possible 2. Immunity against other signals	1. Final speed proved to be slow and cost of products used was high
Advantages	TradeOffs
1. Finished,the project in a short timeframe 2. Testing on the track was smooth	1. Directional control was rudimentary at best

